

Housing Market Trends Analytics Dashboard

Project Documentation

1. Introduction

Project Title: Housing Market Trends Analytics Dashboard

Purpose: The Housing Market Trends Analytics Dashboard is designed to analyze, visualize, and present housing market data through interactive dashboards. The system enables buyers, sellers, investors, and real estate analysts to make informed decisions by providing meaningful insights into housing prices, market trends, and property statistics.

2. Project Overview

The application collects and processes housing market datasets, performs data cleaning and preprocessing using Python, and generates interactive visualizations. The dashboard integrates Tableau reports and web-based charts to present key performance indicators (KPIs), market trends, and comparative analytics.

Key Features:

- Housing market trend analysis
- Interactive dashboards
- Data preprocessing and cleaning
- Tableau visualization integration
- KPI monitoring
- Dynamic charts and graphs

3. System Architecture

Frontend: HTML5, CSS3, JavaScript, Chart.js

Backend: Flask (Python)

Data Processing: Pandas, CSV datasets

Visualization: Tableau, Chart.js

4. Software Requirements

Python 3.x, Flask, Pandas, NumPy, Matplotlib (optional), Chart.js, Visual Studio Code, Tableau Desktop, Git

5. Installation and Setup

1. Install Python 3.x.
2. Create a virtual environment.
3. Install dependencies using `pip install -r requirements.txt`.
4. Run the application using `python app.py`.

6. Project Structure

app.py, templates/, static/, dataset/, tableau/, requirements.txt

7. Application Workflow

Load dataset → Clean and preprocess data → Analyze → Generate charts → Display dashboard → User interaction through filters.

8. API Documentation

Flask routes render dashboard pages, load datasets, provide processed data, and integrate Tableau visualizations.

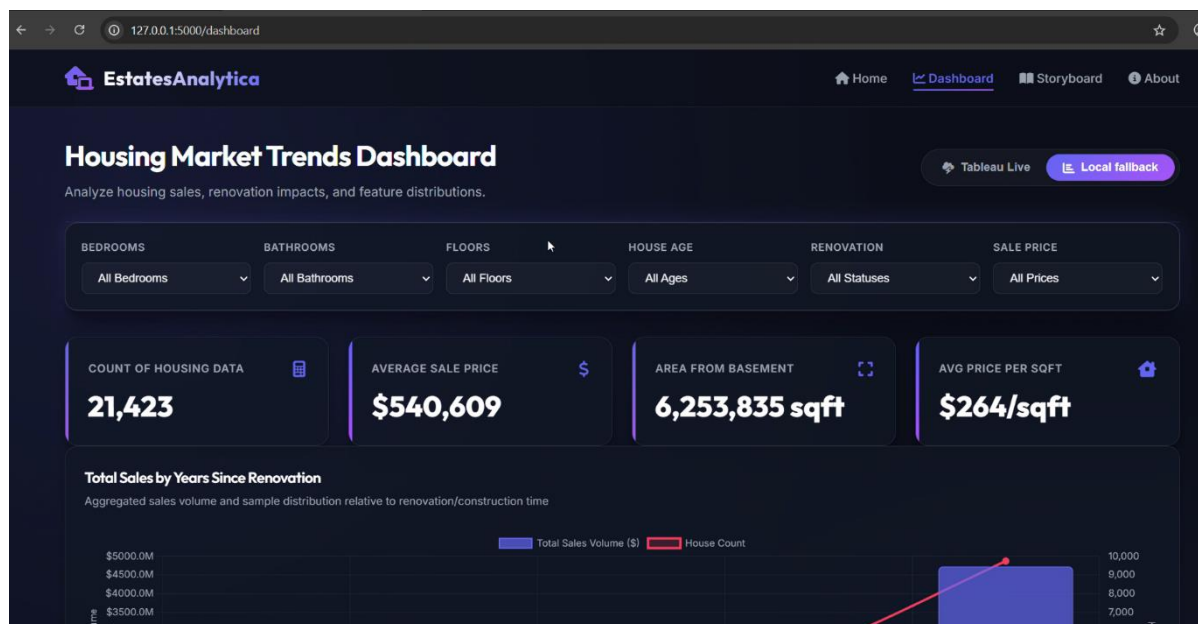
9. User Interface

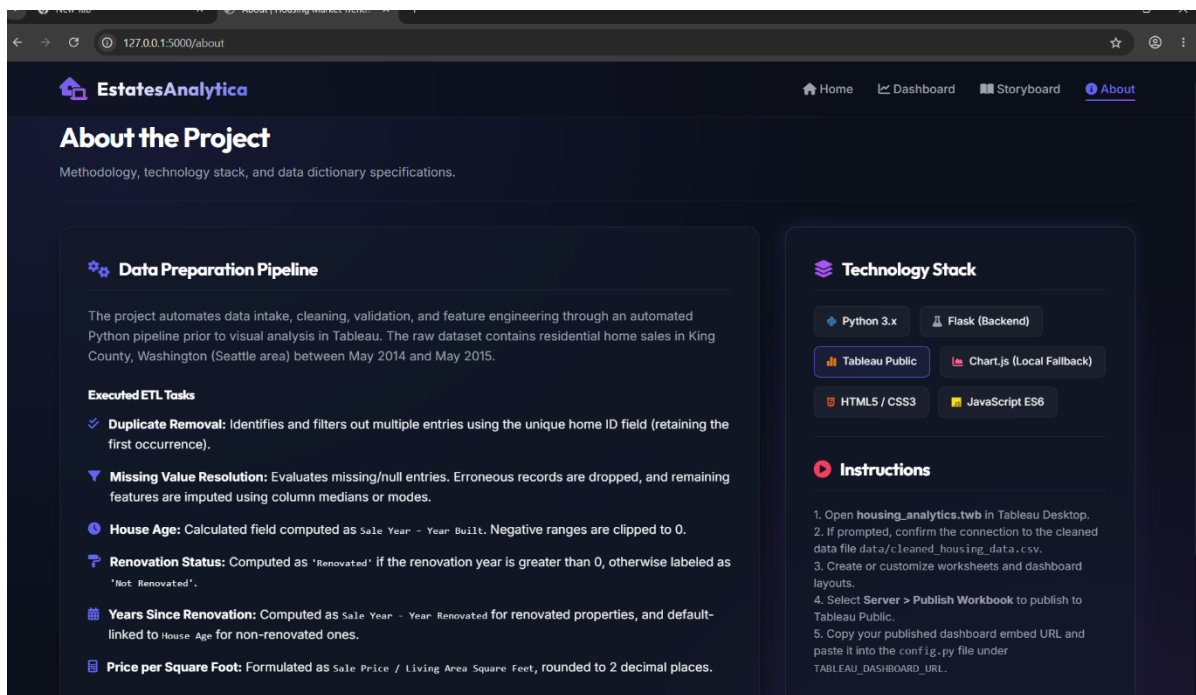
Interactive filters, KPI cards, housing price trends, market comparison charts, Tableau dashboards, and responsive design.

10. Testing

Tested for dataset loading, preprocessing, dashboard rendering, chart generation, filtering, and application performance.

11. Screenshots





12. Known Limitations

Large datasets may increase loading time. Authentication and real-time data integration are not implemented.

13. Future Enhancements

AI-based house price prediction, recommendation engine, authentication, cloud deployment, mobile support, real-time data, advanced analytics, and downloadable reports.

14. Conclusion

The project provides an interactive analytics platform for understanding housing market trends using Flask, Pandas, Tableau, and modern web technologies.